MID-TERM EXAM (GRAPH MINING – SPRING 2026)

Full Name: ___________________________________

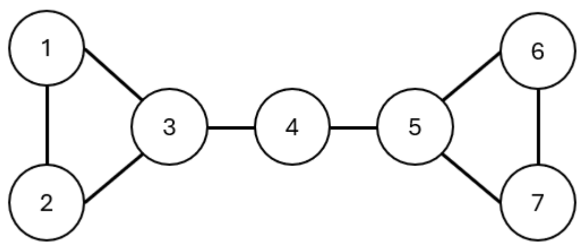
Student ID:___________________________________

Instructions:

- Each response must include both the relevant formula(s) and a clear, step-by-step solution leading to the final answer.
- All answers must be written in English.

Question 1: Degree Centrality and Degree Distribution (10 points)

Consider the undirected graph G with nodes



Given:

- **Unnormalized Degree Centrality**: $C_D(v) = \sum_{j=1}^{|V|} A_{vj}$

  Where:
  - $C_D(v)$is the degree centrality of node v.
  - $\deg(v)$ is the total number of edges incident to node v.
  - $|V|$ (often denoted as N) is the total number of nodes (vertices) in the graph.
  - A is the adjacency matrix of the graph, where $A_{vj} = 1$ if there is an edge between node v and node j, and 0 otherwise.
- **Degree distribution** p_k: $p_k = \frac{N_k}{|V|}$

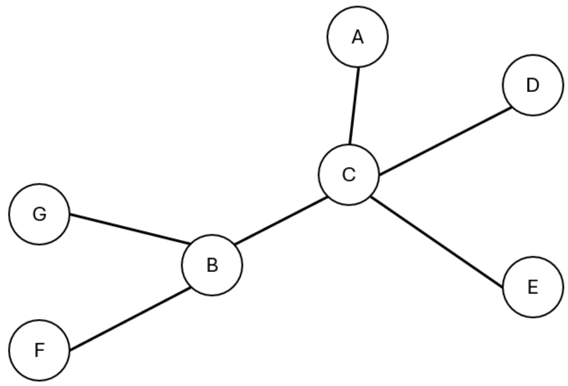
  Where
  - N_k is the total count of nodes with degree k
  - $|V|$ is the total number of nodes in the network.

a) Compute the **<u>Degree centrality</u>** of every node and then compute the **<u>Degree distribution</u>** of the graph. (7 points)

b) One student claims that: *"A node with the highest degree centrality is always the most important node in a network"*. What is your opinion about this statement? (3 points)

Question 2: Betweenness Centrality and Closeness Centrality

Given an undirected graph G of 7 nodes.



Known that:

- **Betweenness centrality** $C_B(v)$:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

  Where:
    - s and t are pairs of nodes in the network.
    - σ_{st} is the total number of shortest paths from the node s to node t.
    - $\sigma_{st}(v)$ is the number of those shortest paths that explicitly pass through the v.
    - If $s = t$, $\sigma_{st} = 1$, and if $v \in \{s, t\}$, $\sigma_{st}(v) = 0$.
- **Closeness Centrality (Normalized)**

$$C'_C(v) = \frac{|V| - 1}{\sum_{u \neq v} d(v, u)}$$

  Where:
    - $d(v, u)$ is the shortest path **distance** between node v and node u.

a) Calculate the **Betweenness Centrality** and **Closeness Centrality** of Node C (7 points)

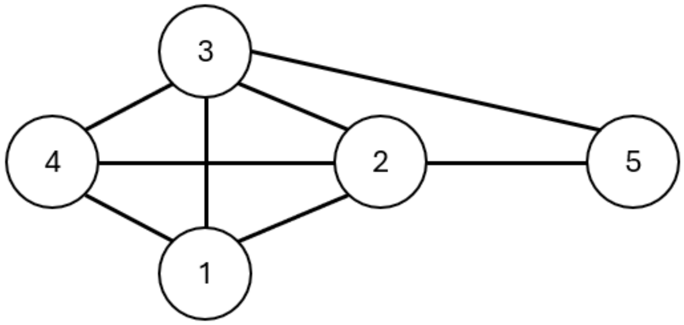
b) If a company wants to identify the employee who:
   - best positioned to connect different departments, or
   - can quickly spread information for the whole organization

   Which centrality measure is most appropriate for each goal? Explain your answer.

   (3 points)

Question 3: Eigen/Katz/PageRank (10 points)

Consider the undirected graph $G = (V, E)$ shown below:



Let the $\alpha = 0.25$ and $\beta = 0.2$

The eigenvalues of the adjacency matrix A of the graph are given as:

$\lambda = \{-1.68, -1.0, -1.0,\ 0.35,\ 3.32\}$

a) Construct the adjacent matrix A corresponding to the graph (4 points)

b) Compute the Katz centrality vector C_{Katz}, defined as:

$$C_{Katz} = \beta(I - \alpha A^T)^{-1} \cdot \mathbf{1}$$

Where $\boldsymbol{I}$ is the Identity matrix and $\mathbf{1}$ is a vector of ones. (3 points)

c) Why does the eigenvector centrality differ from degree centrality? Explain the idea of "being connected to important nodes makes a node important." (3 points)

Question 4: Fruchterman-Reingold Layout (10 points)

Suppose four nodes A, B, C, D lie on a line at positions: $A = (0,0)$, $B = (1,0)$, $C = (3,0)$, $D = (4,0)$ and edges are (A, B) and (C, D). Suppose $k = 2$.

In this problem, please use:

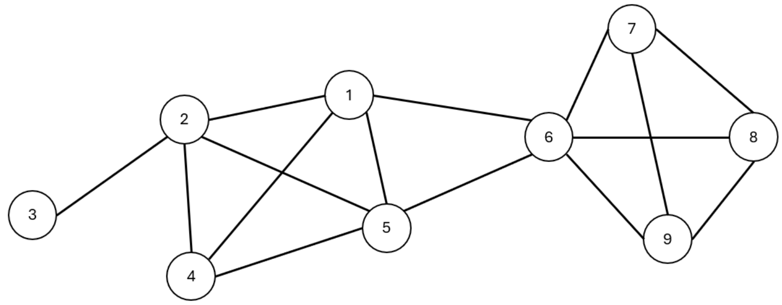
- Attractive force: $f_a(d) = \frac{d^2}{k}$
- Repulsive force: $f_r(d) = \frac{k^2}{d}$

a) <u>**Compute the attractive force**</u> on edge (A, B) and the <u>**repulsive force**</u> between two nodes, B and C. (5 points)

b) What will likely happen if a Fruchterman-Reingold layout is used only with attractive forces and ignores repulsive forces? (5 points)

Question 5: Clustering Coefficient

1. Consider an undirected graph G of nine nodes given in the following figure. (10 points)



a) Calculate the **global clustering** coefficient C_i (4 points)

$$\text{Given: } C_i = \frac{\#of\ closed\ triplets}{\#of\ connected\ triplets}$$

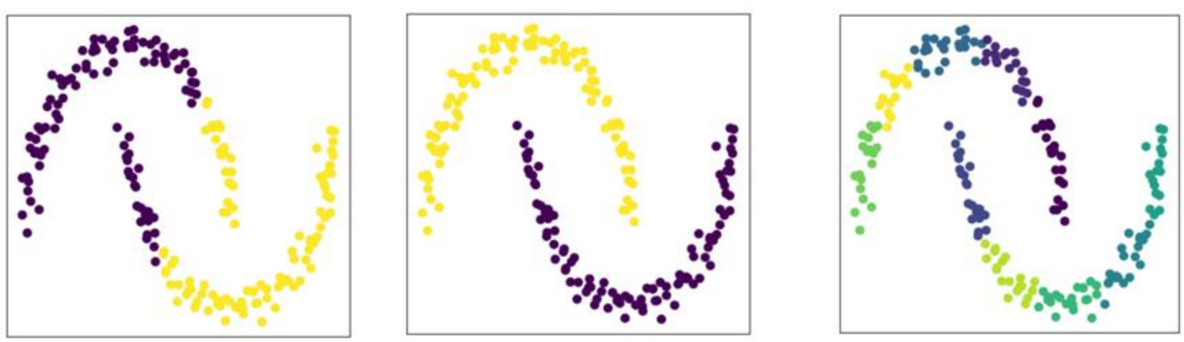
b) Calculate the **average clustering coefficient** $\langle C \rangle$ in the graph G (4 points)

$$\text{Given: } \langle C \rangle = \frac{1}{N} \sum_{i=0}^{N} C_i$$

c) Clustering coefficient is closely related to the presence of triangles in the network. Why are triangles important for understanding local structure? (2 points)

Question 6: Spectral Clustering

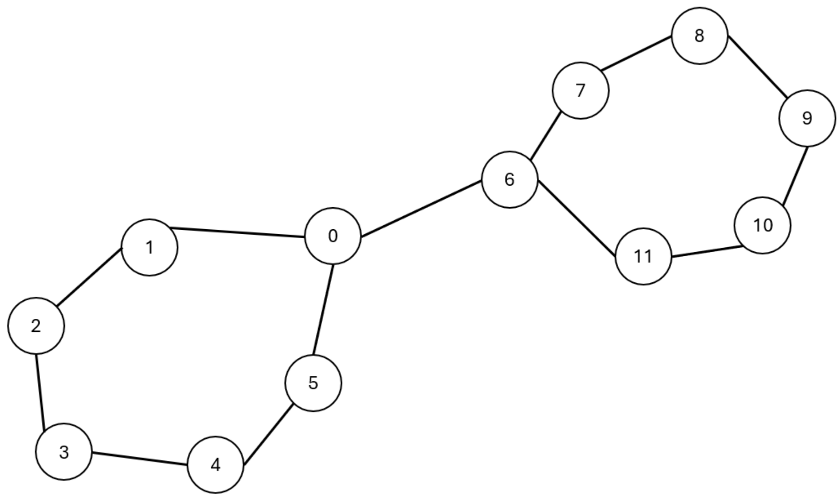
Given three clustering results for two moons, shown in Figures A, B, and C. Which figure most likely represents the output of spectral clustering? **Explain your choice**. (5 points)



A B C

Question 7: Min-Cut and Normalized-Cut (10 points)

Given an undirected graph G of twelve nodes, given as follows. There are two communities that exist in the graph, $A = \{0, 1, 2, 3, 4, 5)$ and $B = \{6, 7, 8, 9, 10, 11\}$



Given:

- **Cut:** A set of edges with only one vertex in a group:

$$cut(A, B) = \sum_{i \in A, j \in B} w_{ij}$$

- **Normalized Cut**

$$NCut(A, B) = \frac{cut(A, B)}{vol(A)} + \frac{cut(A, B)}{vol(B)}$$

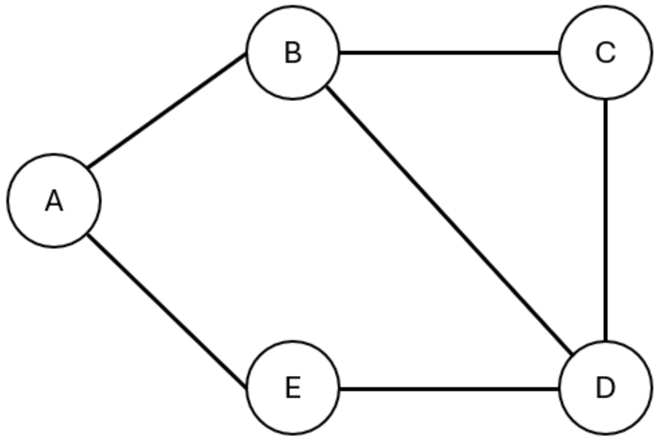
Where:

- $vol(A)$ is the volume of set A, defined as the sum of the degrees of all nodes within A: $vol(A) = \sum_{u \in A} d(u)$ and $d(u)$ is the degree, or sum of edge weights, connected to node u
- $vol(B)$ is the volume of set B

a) **Calculate min-cut** and **normalized cut** measurements of ***A*** and ***B*** (5 points)

b) Why was the idea of **Normalized-cut** introduced? What weakness of Min-cut is N-Cut trying to solve? (5 points)

Question 8: Edge Betweenness (10 points)



Given:

$$C_B(e) = \sum_{s \neq t} \frac{\sigma_{st}(e)}{\sigma_{st}}$$

Where:

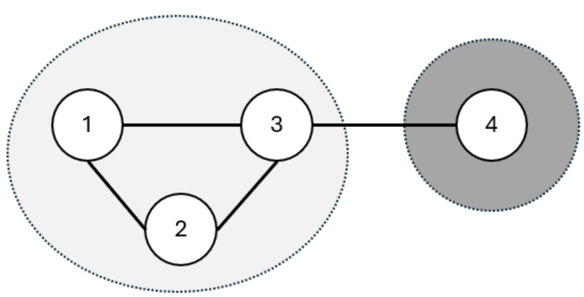
- s and t are pairs of nodes in the graph.
- σ_{st} is the total number of shortest paths from node s to node t.
- $\sigma_{st}(e)$ is the number of those shortest paths that explicitly pass through edge e.

a) Compute the **edge betweenness** of edge (A, B) (7 points)

b) In a transportation network, one road connects two large cities, while many roads lie inside each region. Which types of roads are likely to have higher edge betweenness? Explain your answer in detail. (3 points)

Question 9: Modularity (10 points)

Consider an undirected graph G of four nodes as illustrated in the following figure. Knowing that, there exist two communities $A = \{1, 2, 3\}$ and $B = \{4\}$



Given:

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{d_i d_j}{2m} \right) \cdot \delta(v_i, v_j)$$

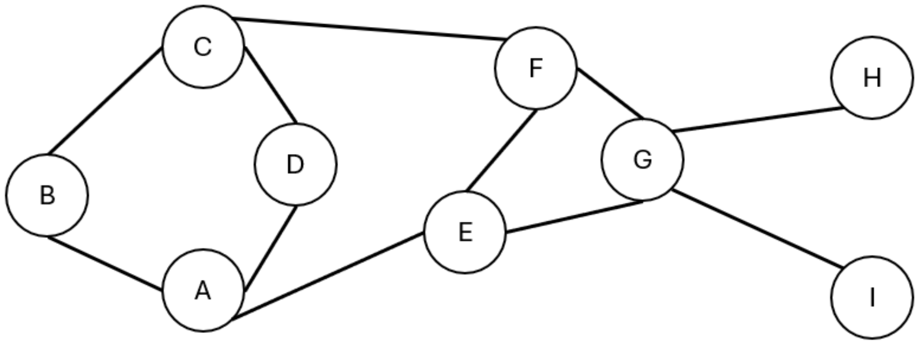
Where:

- $\delta(v_i, v_j) = \begin{cases} 1 & \text{if } v_i \text{ and } v_j \text{ are in the same community.} \\ 0 & \text{otherwise.} \end{cases}$
- m is the number of total edges
- A is the adjacency matrix of G
- d_i is the degree of the node v_i

a) Calculate the **Modularity** of the two communities (6 points)

b) In a social network, modularity is used to evaluate a proposed division of users into communities. What would a high modularity value suggest about the social structure? (4 points)

Use the following graph to answer questions 10 and 11



Question 10: Jaccard Coefficient (5 points)

$$\text{Given: } J(A,B) = \frac{|A \cap B|}{|A \cup B|}$$

a) Calculate the **Jaccard coefficient** of the non-edge pair (A, C) (4 points)
b) Why might the Jaccard Coefficient be more informative than simply counting the number of common neighbors between two nodes? (1 point)

Question 11: Adamic-Adar Index (5 points)

$$\text{Given: } A(u,v) = \sum_{w \in N(u) \cap N(v)} \frac{1}{log|N(w)|}$$

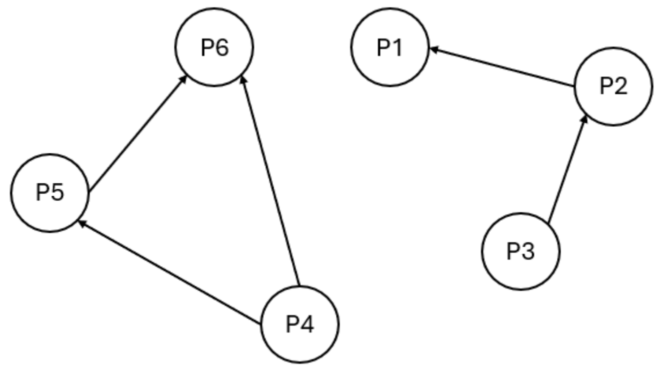
Where:

- $N(u) \cap N(v)$: The common neighbors between the nodes u and v
- w: A specific common neighbor in that set
- $|N(w)|$: The total number of neighbors of the common neighbor w

a) Calculate the **Adamic-Adar Index** of the non-edge pair (C, E) (4 points)
b) What does a high Adamic-Adar Index between two nodes suggest about their relationship in the network? (1 point)

Question 12: Eigen/Katz/PageRank

Consider this mini-internet: (5 points)



Given: **Original PageRank Formula** as:

$$PR(A) = (1 - d) + d \sum_{i=1}^{n} \frac{PR(T_i)}{C(T_i)}$$

Where:

- $PR(A)$: The PageRank of the target page A.
- $T_1, T_2, \ldots T_n$: The pages that have an outbound link pointing to page A.
- $PR(T_i)$: The PageRank score of the linking page T_i.
- $C(T_i)$: The total number of outbound links on page T_i.
- d: The damping factor. (In this assignment, use $\mathbf{d = 0.85}$)

a) **<u>Calculate the PageRank</u>** of all pages for **2 iterations**. Consider the initial value for each node is 1. (4 points)

b) If Katz centrality already improves the eigenvector centrality, why was PageRank still needed? What additional problems does PageRank address? (1 point)